

Benjamin Farsijani

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EDUCATION

University of Ottawa — Honours B.Sc., Computer Science

Sep 2023 – May 2027

GPA: 3.7 / 4.0

Coursework: Probability & Statistics, Linear Algebra, Data Structures & Algorithms, Operating Systems, Computer Architecture, Computer Networks, Systems Programming

TECHNICAL SKILLS

Languages: Python, Go, C/C++, Java, Rust, TypeScript/JS, HTML/CSS, SQL, Bash

ML & Data: PyTorch, XGBoost, scikit-learn, NumPy, pandas, NetworkX; time-series modeling, walk-forward CV, anomaly detection, graph ML, Monte Carlo simulation

Systems & Infra: Linux/POSIX, multithreading, TCP/IP, epoll, Docker, PostgreSQL, SQLite, Flask, REST/WebSocket

Embedded: ESP32 firmware, I2C/SPI/UART, sensor integration, hardware bring-up, 3D printing

PROJECTS

Benny — Algorithmic Crypto Trading System

Production

Python, XGBoost, CCXT, Binance API, PostgreSQL, Docker, Flask

- Built end-to-end ML trading system ingesting real-time market data from Binance across 6 liquid pairs; runs continuously in production with a containerized dashboard tracking equity and trade history.
- Engineered 50+ features with strict zero-lookahead construction: GARCH(1,1) volatility, Parkinson high-low range, VWAP deviation, OBV, ADX, Stochastic RSI, multi-timeframe log returns.
- Trained XGBoost directional classifier validated via 5-fold walk-forward CV; reported per-fold Sharpe, max drawdown, hit rate, and simulated P&L vs buy-and-hold baseline.
- Designed full risk layer: probability-gated entries, 3% TP / 1.5% SL, 0.1% fee modeling, per-symbol sizing across \$6K paper portfolio.
- Extending the system with on-chain and sentiment features, ensemble multi-model voting, and graduated live-capital deployment with reduced sizing for production validation.

Azadi — Threat Network Intelligence Dashboard

In Development

Python, NetworkX, GDEL T, Dash, Cytoscape, SQLite

- Building intelligence platform analyzing actor dynamics during the Iranian Revolution (1979–1981) from GDEL T event data.
- Ingested GDEL T 1.0 bulk data into a directed weighted NetworkX graph; nodes typed by CAMEO actor codes across Government, Military, Religious, and Opposition factions.
- Implemented Louvain community detection and composite influence scoring (PageRank + HITS); daily z-score anomaly detection surfaced 15 event spikes including the Nov. 4, 1979 US Embassy hostage crisis.
- Migrating ingestion to the GDEL T 2.0 real-time stream with temporal community-evolution tracking and an LLM-based event summarization layer over the graph.

Celsius — IoT Environmental Monitoring System

1st Place, Design Day

ESP32, C/C++, React Native, BME680, MAX9814, ML

- Designed and deployed wall-mounted IoT system for Garderie Bernadette Child Care Centre at the University of Ottawa.
- Integrated BME680 (temp/humidity/VOC) and MAX9814 (sound) over I2C with WiFi streaming, 3D-printed lithium-battery-powered enclosure, and React Native app.
- Built ML anomaly-detection layer flagging air-quality dips, noise spikes, and temperature excursions with real-time alerts.

Stochastic Control & MPC for Thermal Oil Recovery

Research

Python, PyTorch, CasADi, NumPy

- Developed stochastic simulation environment modeling delayed nonlinear dynamics (steam injection → reservoir temperature → production output); generated 100k+ synthetic multivariate observations.
- Implemented constrained MPC with a learned PyTorch state-space model solved in CasADi to optimize steam-to-oil ratio against heuristic baselines.
- Validating the controller against historical SAGD field data with ensemble MPC for uncertainty quantification, benchmarked against industry steam-scheduling baselines.

EXPERIENCE

Founder — Benny Blends Barbershop

2018 – Present

- Built and ran a barbershop from a home studio while balancing school full-time; \$25,000+ cumulative revenue (\$15K peak year), thousands of clients served, all operations owned independently.

Technician — TPH Plumbing and Heating

Jun 2024 – Aug 2025 (Summers)

- Worked alongside licensed plumbers and gas technicians on large commercial job sites (crews of 10+, buildings up to 14 stories): cut, threaded, soldered, and fit pipe; supported mechanical room builds and equipment setup.
- Coordinated daily with electricians, HVAC, and GCs to keep work on schedule; followed strict site safety while moving heavy piping and materials across active sites.

Competitive Soccer Coach — Ottawa South United

Mar 2019 – Jun 2024

- Coached 500+ youth athletes across competitive recreational programs over five seasons.